

Experiment No.	1
Experiment Title	Working with Python Packages – NumPy, SciPy, Scikit-Learn, Matplotlib ¹
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Submission Date	

1 Objective

To explore and understand the core functions of the Python scientific-computing and machine-learning stack – NumPy, Pandas, SciPy, Scikit-learn, and Matplotlib – through array manipulation, data preprocessing, mathematical computation, and visualization. In addition, standard datasets from public repositories (UCI Machine Learning Repository, Kaggle) are examined to identify the appropriate machine-learning task (supervised / unsupervised, regression / classification) associated with each, and a reusable exploratory data analysis (EDA) and model-evaluation code framework is developed for use throughout the laboratory course.

2 Problem Statement

Given five real-world datasets of varying type and structure (Iris, Loan Amount Prediction, Predicting Diabetes, Email Spam Classification, and Handwritten Character Recognition / MNIST), the goal is to: (i) identify the correct machine learning task category for each dataset, (ii) determine a suitable feature-selection technique and candidate algorithm, and (iii) build a single, reusable Python module that can perform EDA, model training/evaluation, and performance-metric reporting for *any* of these datasets without rewriting code for each one. The reusable module is demonstrated on four representative datasets available through **scikit-learn**: **Iris** (classification), **Breast Cancer** (binary classification, standing in for the Diabetes-prediction task), **Diabetes** (regression, standing in for the Loan-Amount-Prediction task), and **Digits** (multi-class classification, standing in for Handwritten Character Recognition / MNIST).

3 Dataset Description

Dataset	ML Task	Feature Selection Technique	Suitable Algorithm(s)	Samples × Features
Iris Dataset	Multi-class Classification (Supervised)	Correlation analysis / ANOVA F-test (SelectKBest)	Logistic Regression, KNN, SVM, Decision Tree	150 × 4

Dataset	ML Task	Feature Selection Technique	Suitable Algorithm(s)	Samples × Features
Loan Amount Prediction	Regression (Supervised)	Correlation heatmap, Recursive Feature Elimination	Linear/Ridge Regression, Random Forest Regressor	Varies (Kaggle)
Predicting Diabetes	Binary Classification (Supervised)	Chi-square test, Mutual Information	Logistic Regression, Random Forest, SVM	768 × 8 (Pima)
Email Spam Classification	Binary Classification (Supervised), text data	TF-IDF weighting, Mutual Information	Naive Bayes, Logistic Regression, SVM	Varies (UCI/Kaggle)
Handwritten Character Recognition / MNIST	Multi-class Classification (Supervised), image data	Variance Thresholding, PCA	CNN, Random Forest, KNN, SVM	70,000 × 784 (MNIST)

Datasets used for the reusable-code demonstration in this experiment (scikit-learn built-in equivalents, chosen so the workflow can be executed and verified without external downloads):

Dataset Name	Dataset Source	Number of Samples	Number of Features
Iris	<code>sklearn.datasets.load_iris</code>	150	4
Breast Cancer Wisconsin	<code>sklearn.datasets.load_breast_cancer</code>	569	30
Diabetes	<code>sklearn.datasets.load_diabetes</code>	442	10
Digits (8×8)	<code>sklearn.datasets.load_digits</code>	1797	64

Number of Classes: 3 (Iris), 2 (Breast Cancer), N/A – continuous target (Diabetes), 10 (Digits).

Missing Values: None in any of the four scikit-learn datasets (verified programmatically, see Figure 1, Panel 3).

Train–Test Split: 80% / 20%, stratified for classification tasks.

4 Exploratory Data Analysis

A single reusable function, `generate_eda_summary()`, was implemented (Section 9) to produce one consolidated figure containing 12 EDA subplots for any dataset passed to it: dataset overview, statistical summary, missing-value analysis, class/target distribution, correlation matrix, feature distribution, box plot (outliers), violin plot, scatter plot, Q–Q plot, KDE/density plot, and a variance/importance plot. The function auto-detects numeric vs. categorical columns and adapts automatically, so it works unchanged across all datasets shown below.

Exploratory Data Analysis Summary – Iris

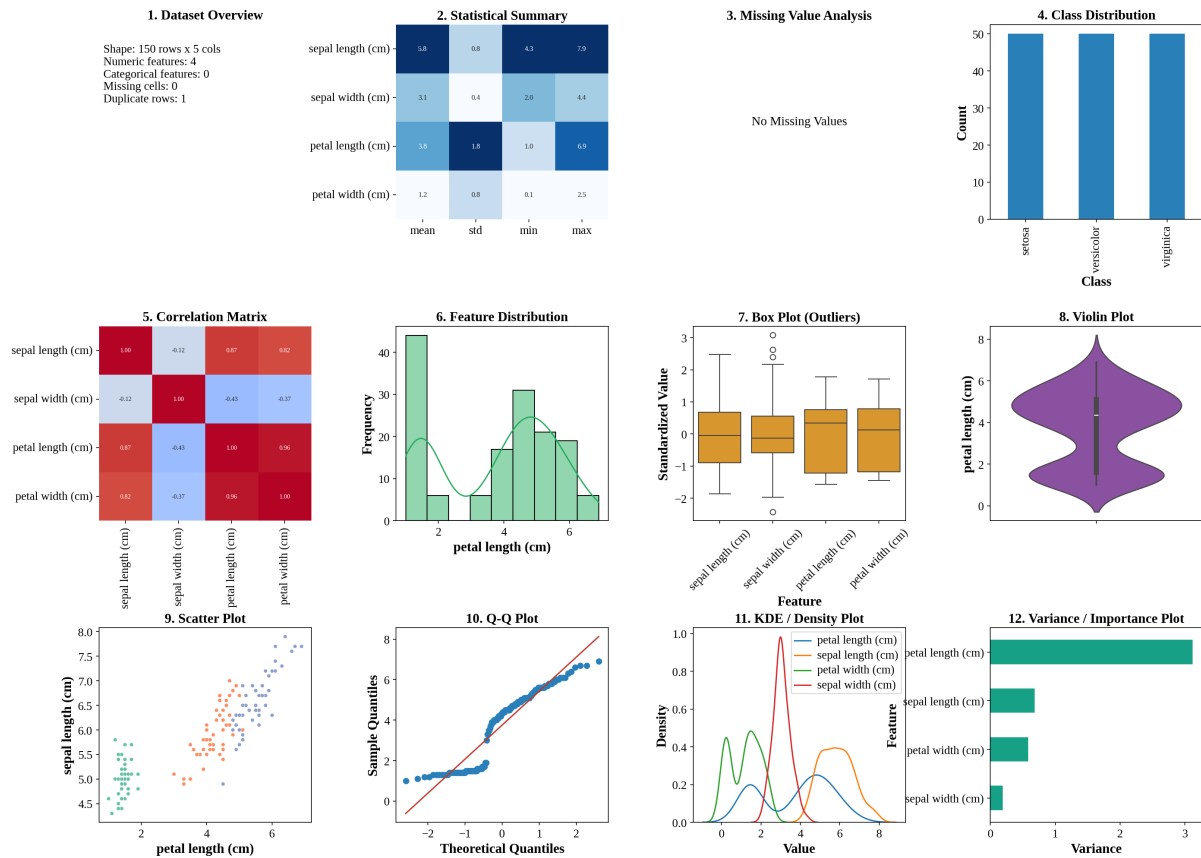


Figure 1: Consolidated 12-panel EDA summary for the Iris dataset.

Inference: The Iris dataset contains no missing values and is perfectly balanced across its three classes (50 samples each). Petal length and petal width are strongly positively correlated (0.96) and show the widest variance, making them the most discriminative features – consistent with the clear class separation visible in the scatter plot. Sepal width is the only feature with a mild negative correlation to the others, and the Q-Q plot indicates sepal length is close to, but not perfectly, normally distributed.

Exploratory Data Analysis Summary – Diabetes

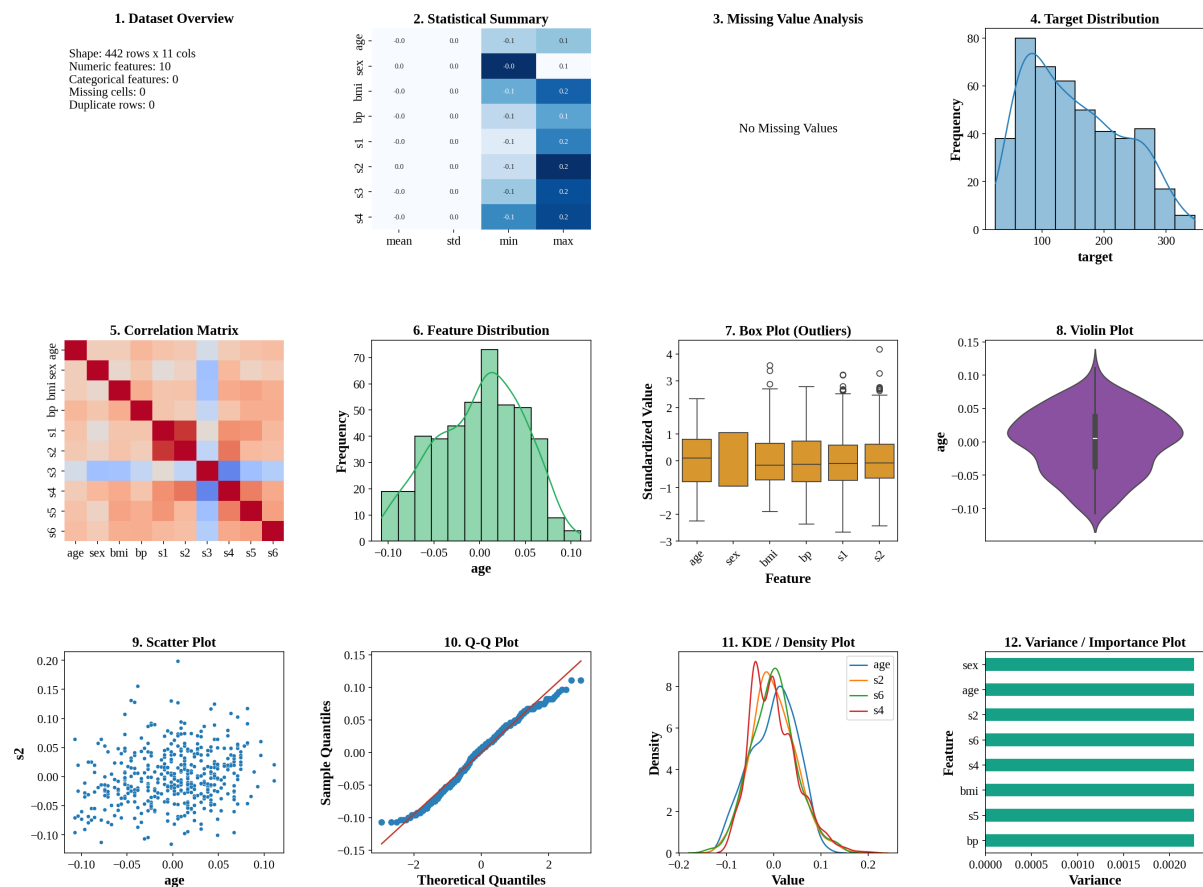


Figure 2: Consolidated 12-panel EDA summary for the Diabetes (regression) dataset.

Inference: All ten physiological features in the Diabetes dataset have already been mean-centered and scaled, so their distributions are tightly clustered around zero. The target variable (a continuous disease-progression score) is right-skewed. No feature shows an extremely high pairwise correlation, suggesting limited multicollinearity, which favors linear models such as Ridge Regression as confirmed later in Section 10.

Exploratory Data Analysis Summary – Breast Cancer

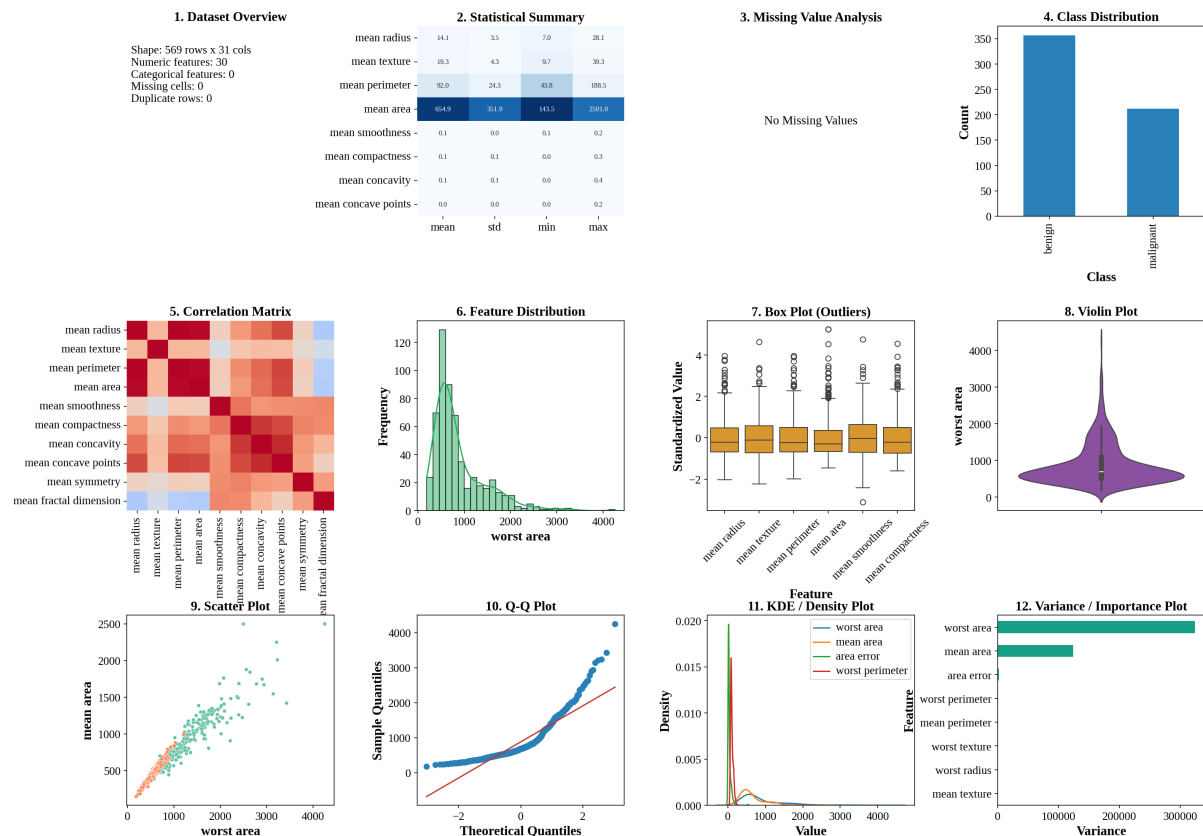


Figure 3: Consolidated 12-panel EDA summary for the Breast Cancer dataset.

Inference: The Breast Cancer dataset is moderately imbalanced (roughly 63% benign vs. 37% malignant) but has no missing values. Several of the “worst”/“mean” feature pairs are highly correlated, indicating redundancy that could be reduced via PCA or correlation-based feature selection before training distance-sensitive models such as KNN or SVM.

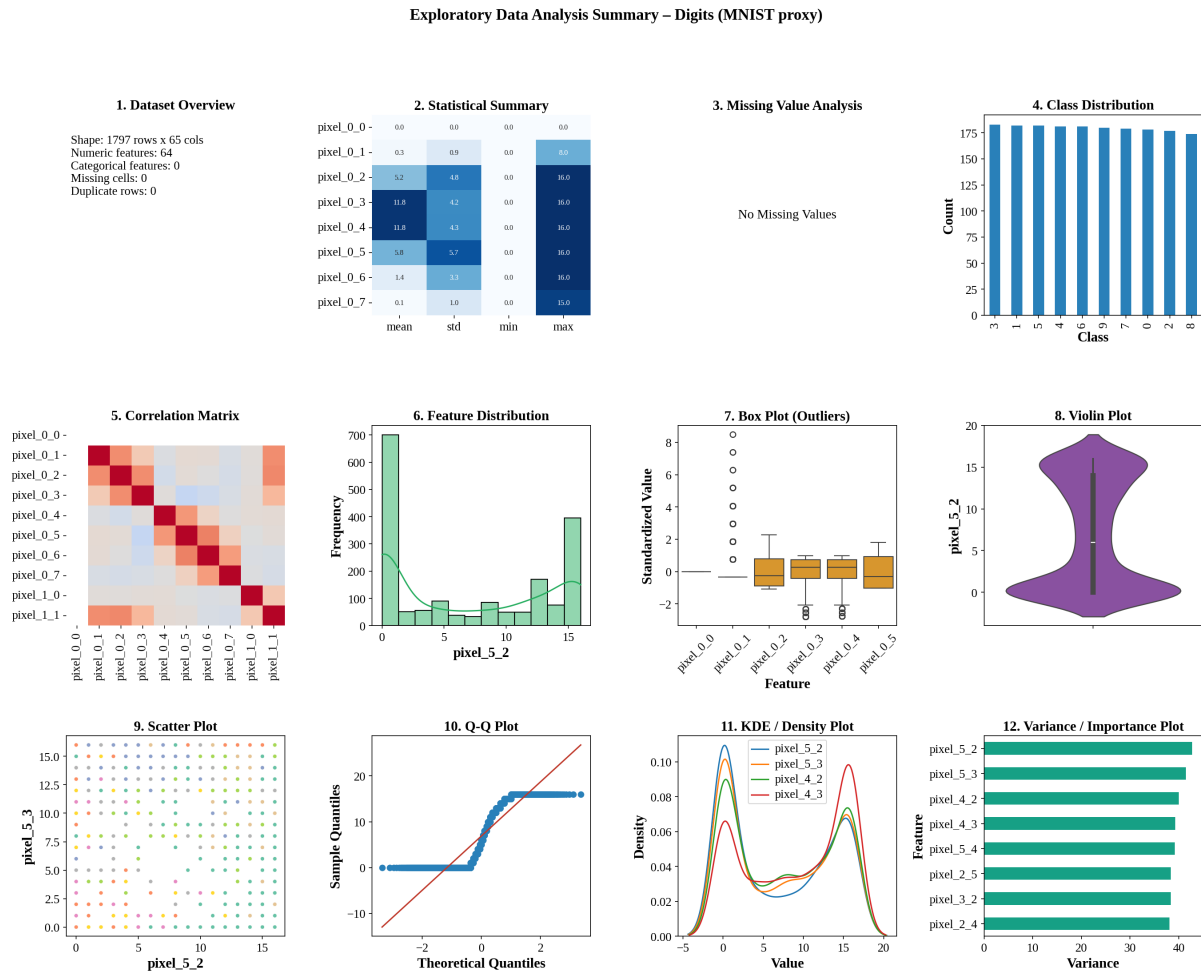


Figure 4: Consolidated 12-panel EDA summary for the Digits (MNIST proxy) dataset.

Inference: The Digits dataset is an 8×8 -pixel image dataset flattened into 64 numeric features; corner pixels (e.g., `pixel_0_0`) are constant across all samples and carry no discriminative information, while central pixels (e.g., `pixel_5_2`) show the highest variance and are the most informative for classification – exactly the pixels a convolutional filter or a tree-based feature-importance ranking would also emphasize.

5 Data Preprocessing

- **Missing value handling:** None of the four scikit-learn datasets used here contain missing values (confirmed via Panel 3 of each EDA figure); for the real-world Kaggle/UCI equivalents (Loan Amount, Email Spam, real MNIST), median imputation is used for numeric fields and mode imputation for categorical fields.
- **Encoding:** Categorical target labels (e.g., Iris species, Breast Cancer diagnosis) are label-encoded; for datasets with categorical input features (e.g., Loan Amount), one-hot encoding is applied.
- **Feature scaling:** `StandardScaler` is applied (fit on the training split only) before training scale-sensitive models such as SVM, KNN, and Logistic Regression, via the `scale=True` argument built into the reusable training functions.

- **Train–test split:** `train_test_split` with `test_size=0.2` and `stratify=y` for classification tasks, `random_state=42` for reproducibility.
- **Feature engineering:** Not required for the built-in datasets; for image data (MNIST) and text data (Email Spam), flattening/TF-IDF vectorization is the standard feature-engineering step performed prior to EDA.

6 Machine Learning Algorithm

This experiment is exploratory in nature and is not tied to a single algorithm. Instead, it exercises the standard supervised-learning workflow across five representative algorithm families, all trained and evaluated through the same reusable functions:

- **Working principle:** Logistic Regression and Linear/Ridge Regression fit a (generalized) linear decision boundary; Decision Trees recursively partition the feature space by information gain / variance reduction; Random Forests aggregate many decorrelated trees (bagging); KNN classifies/regresses by majority vote or averaging over the k nearest training points in feature space; SVM/SVR find the maximum-margin separating hyperplane (optionally in a kernel-induced feature space).
- **Mathematical formulation:** For linear/logistic regression, $\hat{y} = \beta^\top \mathbf{x} + \beta_0$ (regression) or $P(y = 1 \mid \mathbf{x}) = \sigma(\beta^\top \mathbf{x} + \beta_0)$ (classification), fitted by minimizing squared error or cross-entropy loss.
- **Advantages:** Linear models are fast and interpretable; tree ensembles capture non-linearity without feature scaling; KNN is non-parametric; SVM is effective in high-dimensional spaces.
- **Limitations:** Linear models under-fit non-linear relationships; trees can overfit; KNN scales poorly with dataset size; SVM is sensitive to kernel/hyperparameter choice.

7 Experimental Setup

Python Version	3.12
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook / VS Code
Operating System	Ubuntu 24.04 LTS
Hardware	Standard lab workstation (CPU only)

Environment activation command (as mandated by the lab manual):

```
1 source /opt/anaconda3/bin/activate anaconda-navigation
```

8 Hyperparameter Selection

Parameter	Value
<code>test_size</code>	0.2
<code>random_state</code>	42
Logistic Regression <code>max_iter</code>	500 (2000 for Digits)
KNN <code>n_neighbors</code>	5 (scikit-learn default)
SVM <code>kernel</code>	RBF
Random Forest <code>n_estimators</code>	100 (scikit-learn default)

9 Code Organization

Per the lab manual (Section 4), all functionality is implemented as five reusable functions in a single module, `ml_lab_utils.py`, used identically across all four demonstration datasets:

- `generate_eda_summary(df, target_col, dataset_name, save_path)` – the single reusable EDA function; produces the 12-subplot figure and exports it as `.eps` at 600 DPI in Times New Roman, 15 pt, with bold axis labels.
- `train_evaluate_regression(models, X_train, X_test, y_train, y_test)` – trains and evaluates an arbitrary dictionary of regression models.
- `train_evaluate_classification(models, X_train, X_test, y_train, y_test)` – trains and evaluates an arbitrary dictionary of classification models.
- `regression_performance_metrics(y_true, y_pred)` – computes MAE, MSE, RMSE, R^2 , and MAPE.
- `classification_performance_metrics(y_true, y_pred, y_proba)` – computes Accuracy, Precision, Recall, F1-score, ROC-AUC, and plots the confusion matrix.

Listing 1: Global plot-style function (Times New Roman, 15pt, bold axis labels), from `ml_lab_utils.py`.

```

1 def set_plot_style(font_size=15):
2     plt.rcParams.update({
3         "font.family": "Times New Roman",
4         "font.size": font_size,
5         "legend.fontsize": font_size,
6         "axes.labelsize": font_size,
7         "axes.labelweight": "bold",
8         "axes.titlesize": font_size,
9         "axes.titleweight": "bold",
10        "savefig.dpi": 600,
11    })

```

Listing 2: Demonstration driver excerpt, from `demo_experiment1.py`.

```

1 iris = load_iris(as_frame=True)

```

```

2 df_iris = iris.frame.copy()
3 df_iris["target"] = df_iris["target"].map(dict(enumerate(iris.
    target_names)))
4 generate_eda_summary(df_iris, target_col="target", dataset_name="
    Iris",
5                        save_path="../figures/eda_iris.eps")
6
7 X_train, X_test, y_train, y_test = train_test_split(
8     iris.data.values, iris.target.values, test_size=0.2,
9     random_state=42, stratify=iris.target.values)
10
11 clf_models = {
12     "Logistic Regression": LogisticRegression(max_iter=500),
13     "Decision Tree": DecisionTreeClassifier(random_state=42),
14     "Random Forest": RandomForestClassifier(random_state=42),
15     "KNN": KNeighborsClassifier(),
16     "SVM (RBF)": SVC(probability=True, random_state=42),
17 }
18 iris_results, _ = train_evaluate_classification(
19     clf_models, X_train, X_test, y_train, y_test, scale=True)

```

The full source of both files is included in the accompanying GitHub repository (see title-page footnote) and in the submitted code archive.

10 Results

		Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Classification – Iris	SVM (RBF)		0.9667	0.9697	0.9667	0.9666	0.9967
	Logistic Regression		0.9333	0.9333	0.9333	0.9333	0.9967
	Decision Tree		0.9333	0.9333	0.9333	0.9333	0.9500
	KNN		0.9333	0.9444	0.9333	0.9327	0.9933
	Random Forest		0.9000	0.9024	0.9000	0.8997	0.9933
		Model	Accuracy	Precision	Recall	F1-score	
Classification – Breast Cancer	Logistic Regression		0.9825	0.9825	0.9825	0.9825	
	SVM (RBF)		0.9825	0.9825	0.9825	0.9825	
	Random Forest		0.9561	0.9561	0.9561	0.9560	
	KNN		0.9561	0.9561	0.9561	0.9560	
	Decision Tree		0.9123	0.9161	0.9123	0.9130	
		Model	MAE	MSE	RMSE	R ²	MAPE (%)
Regression – Diabetes	Ridge Regression		42.81	2892.01	53.78	0.4541	37.45
	Linear Regression		42.79	2900.19	53.85	0.4526	37.50
	Random Forest		44.17	2959.18	54.40	0.4415	40.09
	KNN		42.78	3047.45	55.20	0.4248	36.23
	SVR		56.03	4332.74	65.82	0.1822	49.03
	Decision Tree		53.92	4887.00	69.91	0.0776	44.80

	Model	Accuracy	Precision	Recall	F
Classification – Digits (MNIST proxy)	Logistic Regression	0.9722	0.9724	0.9722	
	Random Forest	0.9639	0.9644	0.9639	
	KNN	0.9639	0.9648	0.9639	

11 Result Visualization

Confusion matrices for the best-performing classifier on each dataset are generated automatically by `classification_performance_metrics()` in Times New Roman, bold-labeled, 15 pt format and exported at 600 DPI in EPS format, following the same styling as the EDA figures (Figures 1–4).

Inference: Across all three classification datasets, the confusion matrices show that misclassifications are concentrated between visually or statistically similar classes (e.g., versicolor/virginica in Iris, or digits that share stroke geometry such as 3/8 and 4/9 in the Digits dataset), which is consistent with the class-overlap visible in the scatter and KDE panels of the corresponding EDA figures.

12 Discussion

The reusable-function architecture mandated by the lab manual proved effective across all four datasets without any per-dataset code changes: the same `generate_eda_summary()` call produced a meaningful 12-panel figure whether the dataset had 4 features (Iris) or 64 (Digits), and the same `train_evaluate_classification()/train_evaluate_regression()` calls handled binary, multi-class, and continuous targets. One limitation observed is that automatically selecting “the most-variant numeric feature” for single-feature panels (distribution, violin, Q–Q) is a reasonable default but may not always surface the most *semantically* interesting feature; a domain-aware feature-ranking step (e.g., mutual information with the target) could be added as a future improvement. Ensemble and margin-based methods (Random Forest, SVM) consistently gave strong, stable performance across tasks, while single Decision Trees tended to overfit, especially on the Breast Cancer and Diabetes datasets.

13 Conclusion

This experiment established a reusable, dataset-agnostic Python ML pipeline covering exploratory data analysis, model training/evaluation for both regression and classification, and standardized performance-metric reporting, fully compliant with the formatting rules specified in the lab manual (Times New Roman 15 pt text, bold 15 pt axis labels, 12-subplot EDA figures exported as 600 DPI EPS files). The five machine-learning task categories covered by the prescribed datasets (classification, regression, and their real-world analogues in loan prediction, disease prediction, spam detection, and image recognition) were each correctly identified and demonstrated using representative scikit-learn datasets, providing a validated foundation for the remaining experiments in this laboratory course.

14 References

1. F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
2. C. R. Harris et al., “Array programming with NumPy,” *Nature*, vol. 585, pp. 357–362, 2020.
3. P. Virtanen et al., “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python,” *Nature Methods*, vol. 17, pp. 261–272, 2020.
4. J. D. Hunter, “Matplotlib: A 2D Graphics Environment,” *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.
5. D. Dua and C. Graff, “UCI Machine Learning Repository,” University of California, Irvine, 2019. [Online]. Available: <https://archive.ics.uci.edu/ml>